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VedaVenture

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Abstract

The traditional methods of learning arithmetic operations emphasize procedural sophistication and drilling. This leads to a lack of motivation and retention of students. However, with mobile learning and gamification becoming very popular for improving learning motivation by making learning accessible anytime and anywhere, there is still a lack of association with highly efficient computing frameworks. To address this issue, a complete mobile learning app was developed using Android Studio. This app has the potential to transform the mental training of students in math by educating and training them on Vedic Mathematics. Vedic Mathematics uses simple rules and patterns to simplify the process of computing. This makes it highly efficient for fast mental computation. This app integrates these highly efficient methods of mathematics with gamification and mobile learning space. A secure login system for protecting user data and elements of gamification and mlearning space are incorporated. These elements include points, learning levels, rewards, and learning streaks. Finally, with gamification and mLearning space incorporated with highly efficient computing methods, this app offers an incredibly interactive platform for improving learning and computation of students.

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1 Introduction

Mathematics plays a fundamental role in education, forming the basis for problem-solving, logical reasoning, and analytical thinking. However, many learners struggle with traditional arithmetic methods due to their procedural complexity and lack of engagement. Conventional teaching approaches often rely on repetitive practice and memorization, which can lead to reduced motivation and poor retention among students.

In recent years, digital learning technologies have emerged as effective tools to enhance educational experiences. Among these, gamification has gained significant attention for its ability to improve user engagement and motivation by incorporating game design elements such as points, levels, and rewards into non-game contexts [1]. Studies have shown that gamified learning environments can increase participation and enhance learning outcomes, although their effectiveness depends on proper design and user context [2].

Simultaneously, mobile learning (M-learning) has transformed the accessibility of education by enabling learners to access content anytime and anywhere. The widespread adoption of smartphones has made mobile platforms a powerful medium for delivering interactive and personalized learning experiences [3]. Mobile-based systems are particularly suitable for short, engaging, and practice-oriented activities such as arithmetic training.

Another important advancement in educational technology is the development of adaptive learning systems. These systems personalize the learning experience by adjusting content difficulty and feedback based on individual user performance. Personalized gamification has been shown to significantly improve motivation, engagement, and cognitive outcomes compared to generic approaches [4].

In parallel, Vedic Mathematics offers an alternative approach to conventional arithmetic by providing efficient mental calculation techniques based on simple rules and patterns. These methods reduce computational complexity and improve calculation speed and accuracy, making them highly suitable for rapid problem-solving [5]. Despite its advantages, Vedic Mathematics has not been widely integrated into modern digital learning platforms.

Although existing research has explored gamification, mobile learning, and adaptive systems independently, there is limited work that combines all these approaches specifically for mental arithmetic training. Furthermore, most current applications lack integration with efficient computational techniques such as Vedic Mathematics, resulting in missed opportunities to enhance both engagement and performance.

To address these limitations, this project proposes a gamified, adaptive mobile application for learning Vedic Mathematics. The system incorporates interactive challenges, dynamic difficulty adjustment, and reward-based progression to create an engaging and effective learning experience. By combining gamification, personalization, and efficient mathematical techniques, the proposed solution aims to improve both user motivation and computational proficiency.

The main contributions of this work are as follows:

- Development of a gamified mobile application for arithmetic learning.

- Integration of Vedic Mathematics techniques for efficient computation.
- Implementation of adaptive difficulty based on user performance.
- Enhancement of user engagement through streaks, rewards, and interactive challenges.

2 Literature Review

2.1 Gamification in Education

Gamification has also become an effective strategy to increase user interaction and engagement. motivation outside of games. Deterding et al. referred to gamification as the use of non-game elements like points, badges and leaderboards in game design. In order to generate gameful experiences, we make some modifications to them. This initial work established the theoretical foundation of gamified systems but did not exist in practice. In this connection, Hamari et al. reviewed a wide range of empirical work. research on gamification and found that gamification tends to result in better user involvement, interest, and involvement. But the study that was highlighted is that the efficacy of gamification is situation specific and depends on user characteristics. Additionally, Ivankovi c et al created an adaptive and gamified learning system known as CodeyFor coding education. The system included such features as experience point (XP), streaks, leaderboards, and adaptive difficulty. However, their findings are evidence of increased student interactions and better learning results. They assessed that more large-scale validation was required.

Table 1: Comparison of Existing Literature

Author (Year)	Focus Area	Methodology	Key Findings	Limitations
Deterding et al. (2011)	Gamification Theory	Conceptual study	Defined gamification and game elements	No empirical validation
Hamari et al. (2014)	Gamification Review	Literature review	Gamification improves engagement	Context-dependent results
Ivanković et al. (2025)	Gamified Learning System	System development + testing	Improved engagement using XP, streaks	Limited sample size
Xiao & Hew (2024)	Personalized Gamification	Experimental study	Personalization improves motivation	Requires complex user modeling
Crompton (2013)	Mobile Learning	Historical analysis	Enables flexible learning	Lacks implementation focus
Patel & Parekh (2021)	M-Learning Adoption	Survey-based study	High mobile usage in education	Limited scalability discussion
Al-Fahad (2009)	User Perception	Survey analysis	Positive student perception	Device limitations
Träff et al. (2025)	Arithmetic Learning	Longitudinal study	Cognitive + number skills important	Focused on children only
Dowker (2019)	Cognitive Differences	Theoretical + empirical insights	Individual differences affect learning	Not app-based
Nawale et al. (2026)	Vedic Mathematics	Experimental analysis	Faster and accurate calculations	Limited digital integration

2.2 Personalized and Adaptive Learning Systems

The recent studies have emphasized the role of personalization in gamified learning. Xiao and Hew explored the concept of the personalized gamification and contrasted it with the one-size-fits-all strategies of the past era of research

Adaptive learning systems also extend the concept of personalization and dynamically vary content and difficulty in response to user performance. Some other adaptive mechanisms, like the feedback loops and difficulty scaling, were also implemented in the Codey system, which allows the learners to study at their pace

2.3 Mobile Learning (M-Learning)

Mobile learning (M-learning) has transformed the educational landscape by enabling learning anytime and anywhere. Crompton provided a historical overview of mobile learning and highlighted its evolution toward learner-centered education [6]. The study emphasized the importance of accessibility and flexibility offered by mobile platforms.

Patel and Parekh examined the adoption of mobile learning in academic environments and reported that the widespread use of smartphones has significantly increased access to educational resources [7]. Their findings suggest that mobile learning systems must be optimized for usability and engagement to maximize effectiveness.

Additionally, Al-Fahad analyzed students' attitudes toward mobile learning and found that learners generally perceive mobile learning positively due to its convenience and ability to enhance communication [3]. However, challenges such as device limitations and potential distractions were also identified.

2.4 Mathematics Learning and Cognitive Factors

The cognitive aspects of learning mathematics need to be understood to develop effective learning tools. Träff et al. studied mental arithmetic skill development and concluded that both domain-specific number skills and domain-general cognitive abilities contribute to arithmetic skills. Their study also revealed that learning and performing require different cognitive processes.

Dowker studied individual differences in arithmetic abilities and emphasized how arithmetic performance varies significantly from person to person due to cognitive, emotional, and experiential factors.

2.5 Vedic Mathematics and Computational Efficiency

Vedic Mathematics offers an alternative approach to traditional arithmetic methods by using simplified techniques for faster computation. Nawale et al. demonstrated that Vedic Mathematics significantly improves computational speed and accuracy by reducing cognitive load and simplifying calculations [5]. The study also highlighted its applicability in modern domains such as engineering and computer science.

The integration of Vedic Mathematics into modern educational systems has the potential to enhance problem-solving skills and promote efficient mental calculation techniques. However, its adoption in digital learning platforms remains limited, indicating a gap in current research.

Table 2: Feature Comparison of Existing Approaches

System / Study	Gamification	Personalization	Mobile Support	Adaptive Difficulty	Math Focus
Deterding et al.	Yes	No	No	No	No
Hamari et al.	Yes	No	No	No	No
Codey System	Yes	Yes	No	Yes	No
Xiao & Hew	Yes	Yes	Yes	Yes	No
M-Learning Studies	No	No	Yes	No	No
Arithmetic Studies	No	Yes	No	No	Yes
Vedic Math Study	No	No	No	No	Yes
Proposed System	Yes	Yes	Yes	Yes	Yes

2.6 Summary and Research Gap

According to the literature, gamification, personalization, and mobile learning are highly effective in improving the outcome of education. Although gamification can be used for better engagement and motivation, personalization can be used for better results by considering the differences between individuals. Mobile learning can also be used for better accessibility, making it a suitable option for modern education.

However, existing systems are mostly based on general gamification techniques, and efficient mathematical techniques such as Vedic Mathematics are not considered in these systems. Moreover, research on adaptive gamification with mobile devices for mental arithmetic training is still in its infancy. This project proposes a gamified adaptive mobile application that uses Vedic Mathematics for better computational efficiency and engagement.

Table 3: Identified Research Gaps

Existing Limitation	Proposed Solution
One-size-fits-all gamification	Personalized gamification using user preferences
Lack of mobile-first systems	Develop mobile-based learning application
No integration of Vedic Mathematics	Incorporate Vedic techniques for fast calculation
Limited adaptive difficulty systems	Dynamic difficulty adjustment based on performance
Lack of engagement tracking	Use streaks, XP, and rewards system

3 Objectives

The primary objectives of VedaVenture are as follows:

1. Develop a user-centric mobile application that teaches Vedic Mathematics through structured, sequential, and gamified modules accessible on Android devices.
2. Implement a dual-mode learning system consisting of Career Mode for guided conceptual learning and Challenge Mode for competitive, time-bound practice.
3. Integrate an OCR engine capable of scanning real-world arithmetic problems, detecting the relevant Vedic Sutra, and providing step-by-step resolution guidance to the user.

4. Design an adaptive difficulty mechanism that dynamically adjusts question complexity based on individual user performance metrics including accuracy and response time.
5. Implement a comprehensive gamification and engagement system featuring experience points (XP), daily streaks, achievement badges, and level progression to sustain long-term user motivation.
6. Provide a sequential Sutra unlocking mechanism that ensures conceptual mastery before progression, preventing learners from skipping foundational techniques.
7. Implement a hint system within Career Mode practice questions to scaffold learning and reduce cognitive overload for users encountering difficulty.
8. Leverage Firebase Authentication and Cloud Firestore for secure user account management and real-time cross-device synchronization of user progress data.
9. Create an accessible and inclusive mobile interface using Material Design principles, supporting multiple screen sizes and providing an intuitive user experience.
10. Contribute to the broader field of educational technology by demonstrating the viability of combining ancient mathematical traditions with modern gamified mobile learning platforms.

4 Methodology

4.1 System Overview

The proposed system is a mobile-based Vedic Mathematics learning system, which can improve users' arithmetic skills. The system is based on a modular and service-oriented architecture, using an Android-based front-end and Firebase Authentication and Cloud Firestore for the backend.

The system is composed of a user interaction component, learning and challenge parts, common computational parts, and a cloud-based database for storage.

4.2 System Architecture

The system architecture is composed of three main layers. The front-end client, which is based on Android, is responsible for user interaction, providing real-time feedback, and using a joystick for input. The second layer is composed of the functional layer, which contains the Practice Module and Challenge Module. The backend layer utilizes Firebase Authentication for user management and Firestore for storage.

4.3 Overall System Workflow

As depicted in Figure 2, the system starts from user authentication through the login or registration process. Once the user successfully authenticates the account, the system will redirect the user to the dashboard where the user can select between the Practice Module and the Challenge Module. Other features are also incorporated into the system.

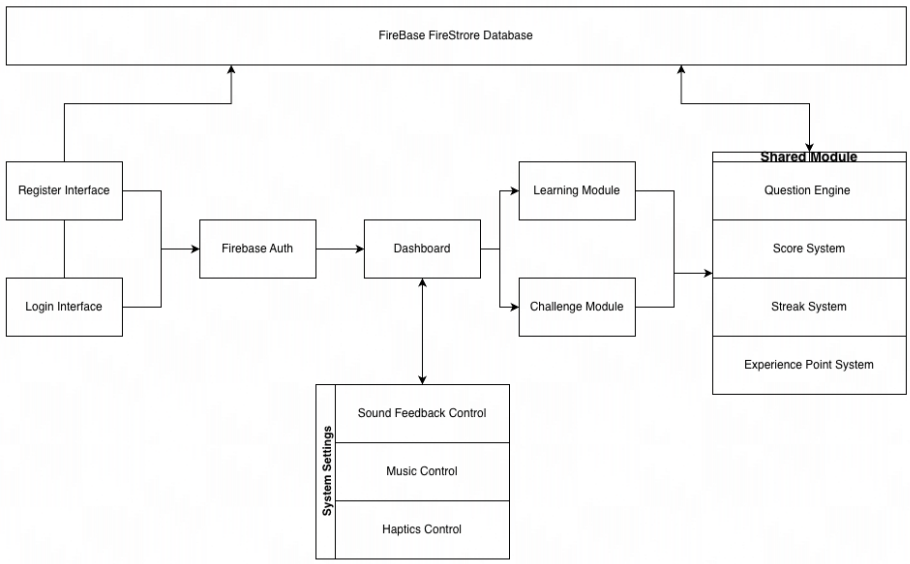


Fig. 1: System Architecture of the Proposed Application

4.4 Practice Module Workflow

The Practice Module is for conceptual understanding of Vedic Sutras. The workflow of this system starts from choosing a Sutra based on user progress. The system will demonstrate, and then users can solve step-by-step problems. The response of users will be checked, and feedback will be generated. If it is wrong, users can try again. After correct completion, it will be updated in the database.

4.5 Challenge Module Workflow

The Challenge Module uses gamification through time-bound problem solving. The questions are generated dynamically depending on the difficulty level. The user input is taken through a joystick. The accuracy of the input and the updating of the score, experience points, and streak are checked. The process continues until the time expires. Then, the results are computed and saved.

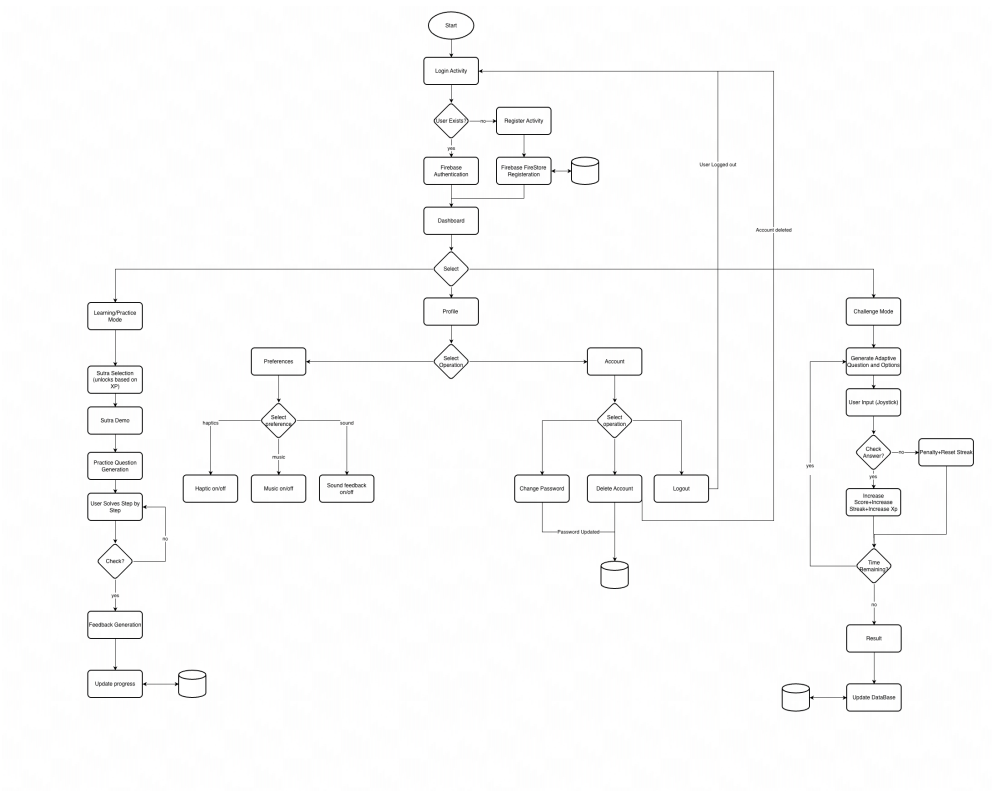


Fig. 2: Overall System Workflow of the Proposed Vedic Mathematics Application

4.6 Adaptive Difficulty Mechanism

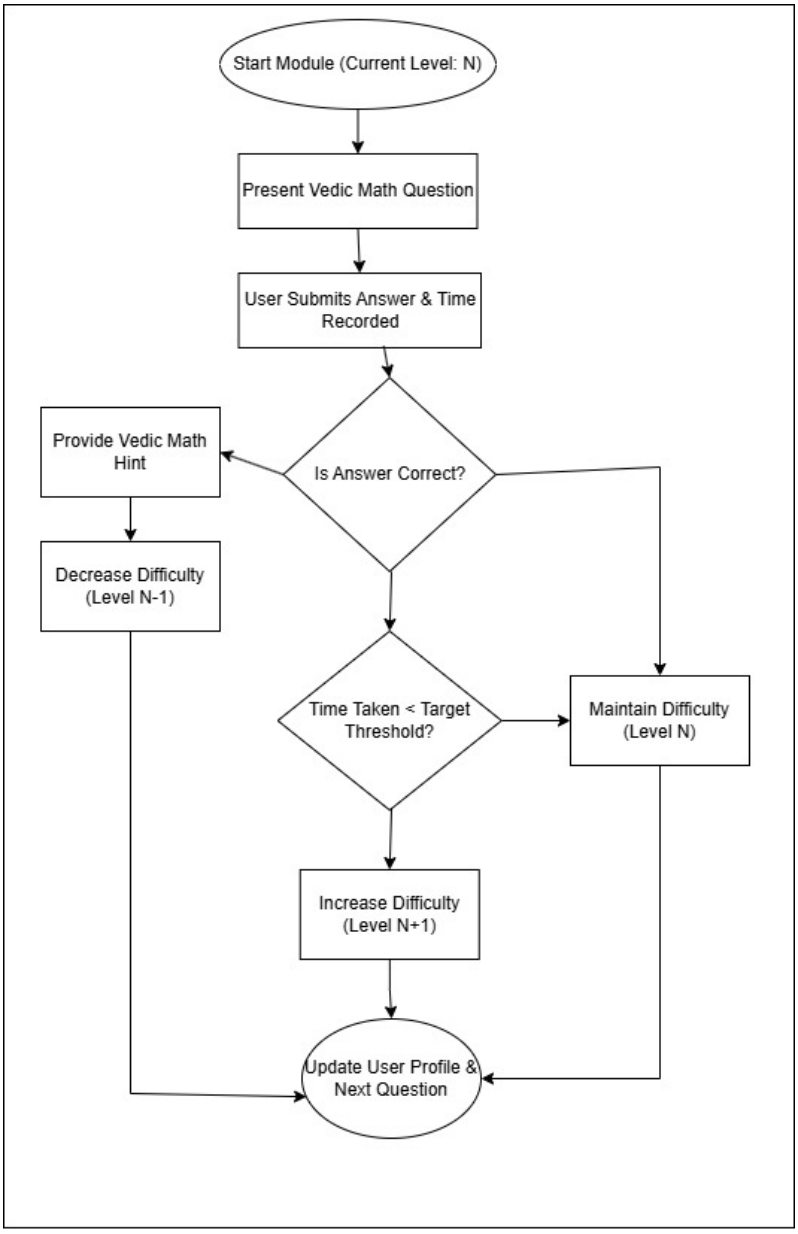


Fig. 3: Adaptive Difficulty Adjustment Mechanism

It also has an adaptive difficulty level that depends on the user's performance, as illustrated in Figure 3. The difficulty level increases if the user correctly answers the question within the desired time, remains the same if the answer is correct but given at a slower rate, and decreases if the answer is incorrect.

4.7 Question Generation Strategy

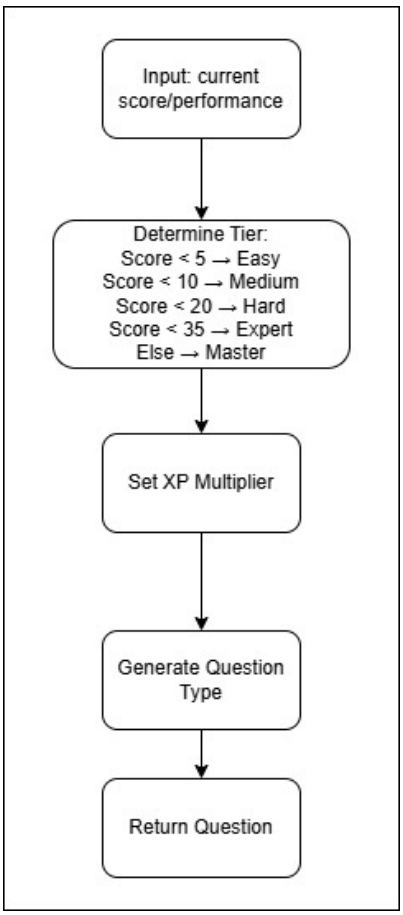


Fig. 4: Question Generation Process Based on Difficulty Levels

The question generation module identifies the difficulty tier depending on the user's score, as shown in Figure 4. Each difficulty tier has a corresponding XP multiplier and question type. Vedic arithmetic questions are generated according to this.

4.8 Joystick-Based Input Handling

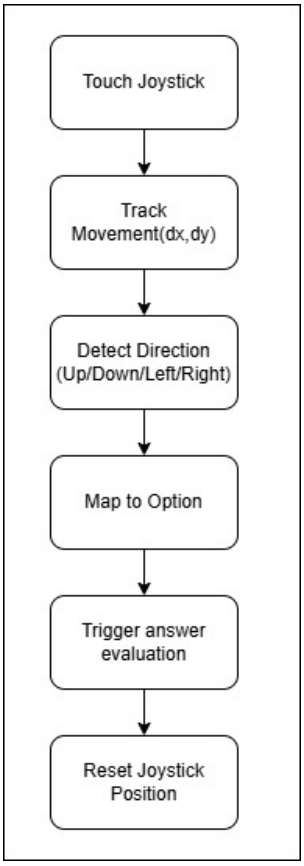


Fig. 5: Joystick Input Processing Mechanism

It (5) uses a joystick input mechanism to improve interactivity. The input mechanism works as follows. The user moves their touch input, and this is translated into a direction. The directions are used as answer choices. The answer choice is then evaluated, and the joystick returns to its original position.

4.9 XP and Streak Management

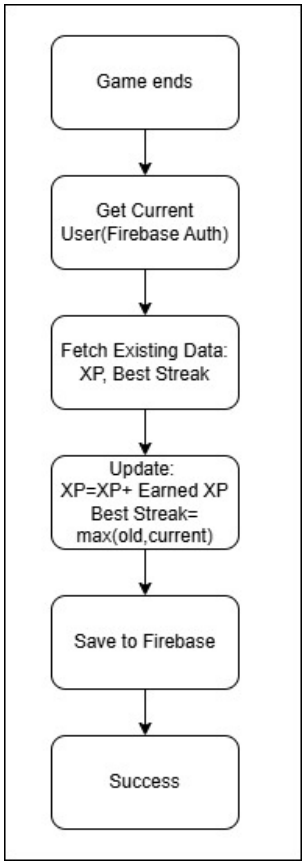


Fig. 6: XP and Best Streak Update Process

The gamification system keeps track of experience points (XP) and the best streak for the user. At the end of each session, it retrieves all the data from Firestore for the user, as shown in Figure 6. The experience points are incremented by the earned value, and the streak is updated as the maximum of the previous best streak and the new streak. The new values are stored.

4.10 Data Persistence and Cloud Integration

Firebase Firestore is used for real-time storage and synchronization of data. The database stores user credentials, XP, streaks, and performance information. Firebase Authentication is used for secure management of user accounts. This approach offers scalability, reliability, and simplicity in backend management.

4.11 System Integration

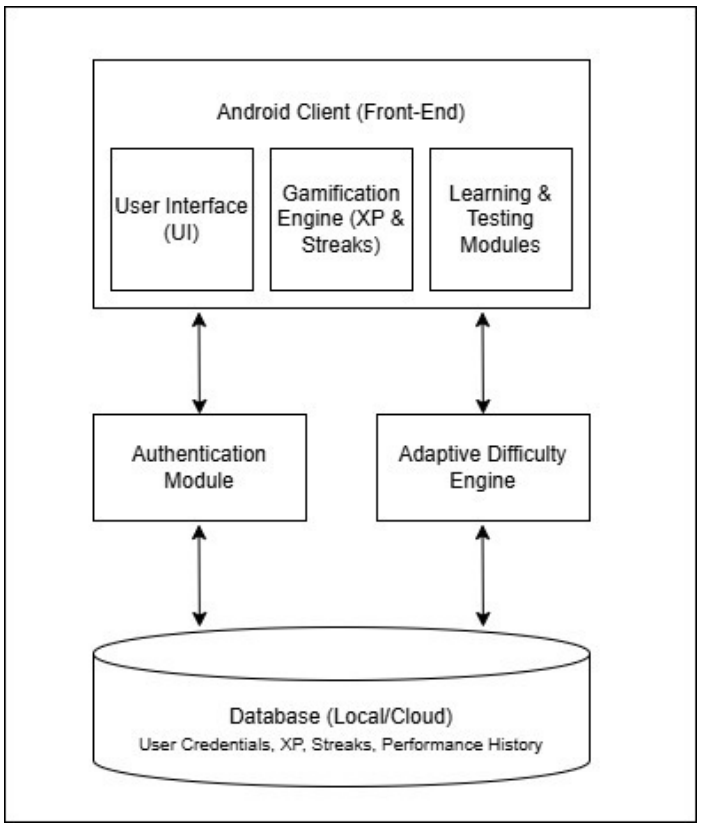


Fig. 7: Abstracted System Integration Model

As a whole, the system comprises different components, which are the user interface, adaptive difficulty, gamification, and database (firebase firestore), as shown in Figure 7.

Presentation Layer (Front-End): In this layer, the user interface was designed using Extensible Markup Language (XML) for faster response and better accessibility for different screen sizes of the mobile devices used by the users. The activities of the android are responsible for handling the main states of the application, which are login and dashboard. In addition, the application uses a fragment for content switching while a user is either learning or testing, as opposed to reloading the screen.

Application Logic Layer (Middleware): In this layer, the logic of the application was designed using Java, based on ViewModel and LiveData of the android operating system, for observing the change of data in real-time, ensuring that as soon as the user's scores or the number of wins in a row are updated in the database, they are reflected on the user interface immediately.

Data Management and Security: To manage user credentials and personalized data in a secure manner, the system utilizes Firebase Authentication for user verification. The passwords and credentials are managed securely by the Firebase Authentication mechanisms. To manage the data, the system utilizes the Cloud Firestore database for storing user-related data, including experience points (XP), streaks, and performance metrics. The data is stored in the cloud, allowing real-time synchronization of the data, thereby eliminating the need to store the data locally using storage mechanisms.

4.12 OCR Scanner Module

One of the most distinctive features of VedaVenture is its OCR (Optical Character Recognition) scanner. This module allows users to point their device camera at any printed or handwritten arithmetic problem and have the app automatically recognize the question, identify the most appropriate Vedic Sutra for solving it, and display a complete step-by-step solution. The OCR pipeline operates as follows: the user activates the scanner from the Dashboard or any relevant screen. The device camera feed is displayed in real time within the app. The user captures an image of the arithmetic problem. The OCR engine processes the image to extract the textual content of the mathematical expression. A pattern-matching and classification module then analyzes the extracted expression to determine its type (multiplication, squaring, addition near a base, etc.) and maps it to the most suitable Vedic Sutra. The solution is then displayed step-by-step using the same guided instructional format used in Career Mode. This feature bridges the gap between classroom learning and everyday life, allowing students to use VedaVenture as a reference tool for homework, exam preparation, or general arithmetic problem-solving. It also increases the utility of the application beyond in-app practice by extending it to real-world problem contexts.

4.13 Integration of Vedic Mathematics Modules

The educational part of the app involves digitization of traditional Vedic math formulas and rules, such as Urdhva Tiryakbhyam (Vertically and Crosswise) and Nikhilam Navatashcaramam Dashatah (All from 9 and the last from 10), which simplify complex multiplication and squaring. Each learning module has two distinct phases:

Instructional Phase: Users are exposed to a new Vedic math rule with visual and stepwise depiction of the computation process. This reduces cognitive overload for learning new techniques.

Application Phase: Users get to engage in interactive and time-constrained activities to apply their learning of these techniques, thereby bridging the gap between learning and fast mental calculation.

Each Vedic Sutra is digitized into a structured learning module within the Career Mode. The digitization process involves breaking down the Sutra's rule into a formal algorithm with clearly defined steps. Each step is paired with a visual demonstration using worked numeric examples. The instructional flow is designed to first present the rule abstractly, then demonstrate it concretely, and finally challenge the user to apply it independently.

4.14 Adaptive Difficulty Algorithm

In order to cater to the various learning needs and maintain the optimal state of engagement, a dynamic adaptive difficulty engine has been engineered. The user performance is assessed on two key factors: accuracy and response time.

The performance score for the user is computed by the engine after the user completes the question. In the case where the user is able to solve the problem correctly within a certain threshold (for example, 80% accuracy and an average response time of ≤ 5 seconds), the difficulty is increased by providing the user with more digits and/or applying advanced Vedic math techniques.

On the contrary, if the user's accuracy is less than a certain threshold, the difficulty is decreased by providing the user with certain hints and/or reviewing the relevant Vedic Sutra.

4.15 Gamification and Retention Engine

In an effort to compensate for the lack of motivation that is normally present in the traditional arithmetic drilling approach, the entire gamification system was incorporated.

Experience Points (XP) and Level Progression: The user is rewarded with various amounts of XP depending on the pace and accuracy of their response. By accumulating XP points, the user can move through various levels of the entire learning curriculum.

Daily Streak: A system was implemented to encourage the user to develop the habit of learning every day. By completing at least one mathematical problem, the user can increase the daily streak counter. A lack of activity for 24 hours resets the counter to zero.

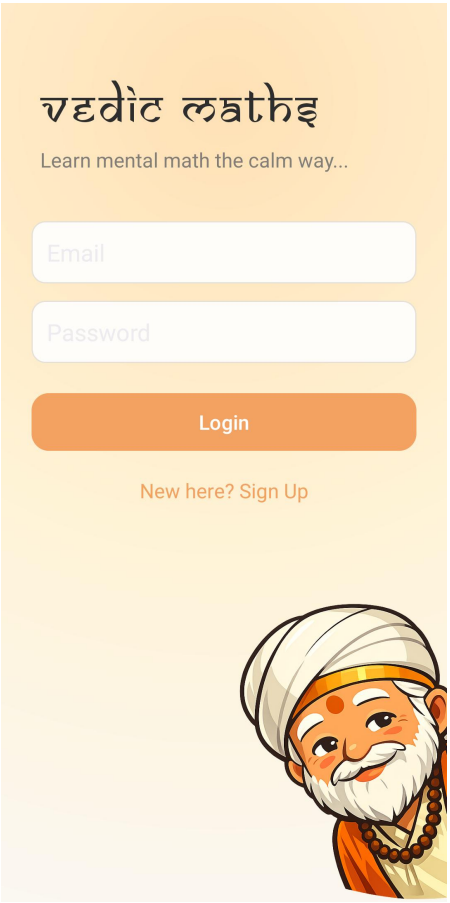
Reward System: The user is rewarded by receiving badges conditionally upon the completion of certain milestones, such as "Master of Multiplication" and "7-Day Streak".

5 Results

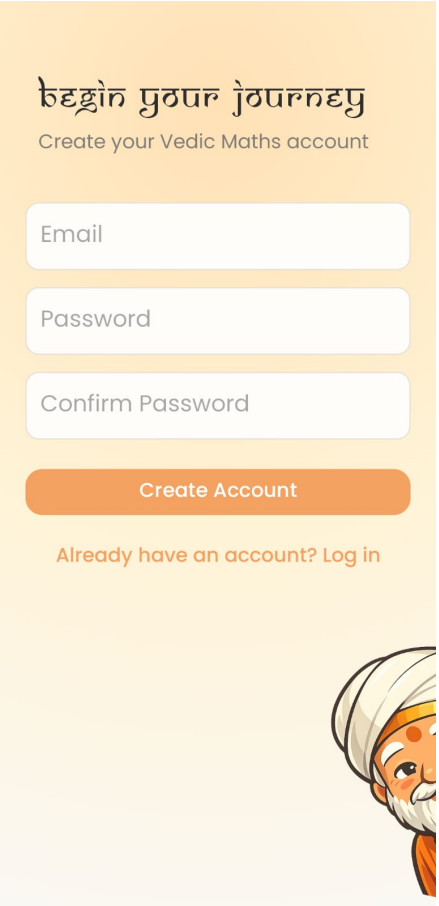
The application that we developed presents a clean, Material Design-compliant interface that guides wonderfully users through their learning journey. The following sections describe the key screens and their roles in the user experience.

5.1 Login and Registration

The Login screen allows existing users to enter their registered email and password to access the application. In case of invalid username or password, appropriate error messages are displayed to the user. Firebase Authentication handles credential verification, and a successful registration redirects the user directly to the Dashboard. Upon successful authentication, users are redirected to the main Dashboard, where they can continue their learning journey from their saved progress.



(a) Login Screen



(b) Registration Screen

Fig. 8: Authentication Screens

5.2 Dashboard

The primary navigation page of the this app is called the Dashboard, which links users to quickly check their learning accomplishments, core modules and profile functionalities. The dashboard is visually designed to have an engaging yet minimal look to illustrate the educational/gamified nature of the application.

On the upper part of the dashboard screen, there are key performance indicators that display the user's total experience points (XP) and the user's best streak. The purpose of these indicators is to encourage practice consistency and allow the user to track their long-term level of accomplishment. There are also quick access icons that provide immediate access to the assorted functions including the OCR scanner and personal profile/setting

The center area of the dashboard will feature the theme branding of the application, along with a character depiction, which will provide a bright and compelling identity to the user interacting with the application. This design will enhance user input and allow for more recurrent use of the application, particularly for younger learners.

Two primary action buttons are presented prominently:

- **Continue Journey:** Redirects the learner to Career Mode, where they can continue structured lessons, unlock new Sutras, and progress through guided learning modules.
- **Challenge Mode:** Opens the competitive practice environment where users solve timed arithmetic problems based on their current skill level to earn XP and improve streaks.

The dashboard is intentionally optimized for fast navigation, ensuring that users can begin learning or practicing within a single tap. It also serves as a personalized home screen by continuously updating user progress data through Cloud Firestore synchronization.

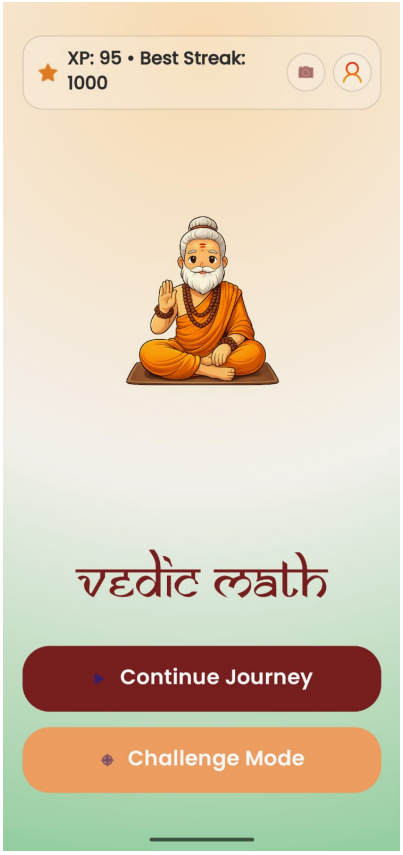


Fig. 9: Dashboard Screen

5.3 Settings and Profile Page

The Settings and Profile Page provides users with a personalized space to manage their account information, monitor achievements, and customize their in-app experience. This screen combines profile analytics with preference controls, making it an important component for both usability and long-term engagement.

At the top of the page, the user's profile section displays the registered email address, avatar, and accumulated Total XP. Presenting this information prominently reinforces a sense of ownership and progression, encouraging learners to remain active on the platform.

The **Preferences** section allows users to control interactive and sensory features of the application through toggle switches. These options improve accessibility and personalization by enabling users to tailor the app environment according to their comfort.

- **Haptic Feedback:** Enables vibration responses during button presses and gameplay interactions, creating a more tactile experience.
- **Background Music:** Allows users to turn ambient music on or off while navigating lessons and practice modules.
- **Sound Effects:** Controls audio cues used for correct answers, incorrect responses, rewards, and navigation events.

The **Account Actions**section includes useful features for managing accounts. Users will be able to securely change their passwords by using Firebase Authentication services, allowing them to keep their accounts safe from the passing of time. There are also options for signing out that can safely terminate sessions on both public and private devices.

The Section has been designed using a card-based layout with a clean, simple, and consistent typographic hierarchy; these design elements have been chosen based off of common design standards found in modern Material Designs.

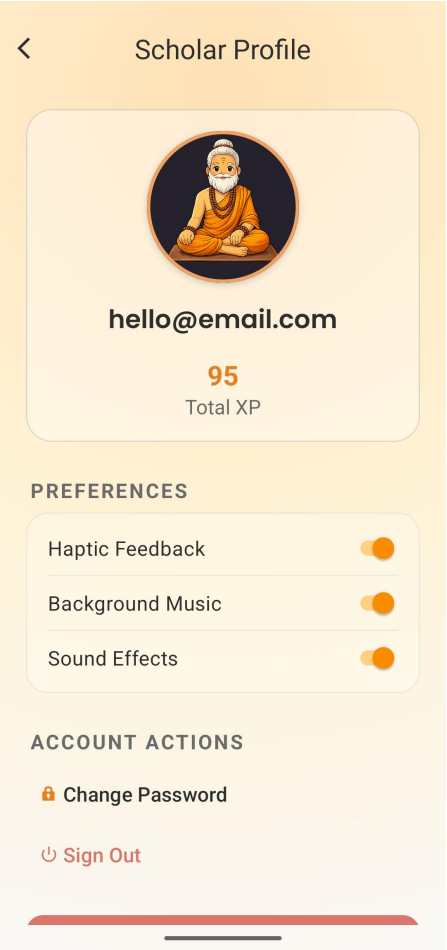


Fig. 10: Settings and Profile Page of VedaVenture

5.4 Career Mode Page

The Career Mode Page serves as the primary guided learning environment of VedaVenture, where users progress through the complete curriculum of Vedic Mathematics in a structured and achievement-oriented manner. It is specifically designed to transform learning into a journey of mastery rather than a collection of isolated lessons.

A distinctive feature of this page is the **customized interactive dial** containing all 16 principal Vedic Sutras arranged in a circular progression interface. Each numbered node on the dial represents one Sutra, creating a visually intuitive roadmap of the learner's advancement. The circular layout symbolizes continuous growth and encourages users to move progressively through the system.

The unlocking mechanism is based on both completion and consistent practice. Initially, only the first Sutra is available to the learner. Subsequent Sutras remain

locked until the previous module has been successfully completed, ensuring that foundational concepts are mastered before introducing more advanced techniques. This prerequisite-based progression prevents random skipping and preserves the logical learning sequence of Vedic Mathematics.

Each Sutra module contains three major stages:

- **Concept Introduction:** The meaning, purpose, and practical application of the Sutra are explained in simplified language.
- **Step-by-Step Learning:** Worked examples and guided demonstrations show how the Sutra is applied to real arithmetic problems.
- **Assessment and Unlocking:** Users solve practice questions based on the lesson. Successful completion unlocks the next Sutra on the dial.

On this page you'll see real-time statistic information about the Student with Dynamic Learner Statistics like Total XP, Completion Rate and Current Curriculum progress. These visual progress indicators will encourage students to continue learning by showing them the progress they've made or accolades they will earn.

Also, since the progress for the student is stored through Cloud Firebase, the student will be able to continue their path through the programme without having to worry about losing any modules or earned rewards when they log in to this application after a period of absences.

In summary, the Career Mode page brings together the structure of a curriculum and gamification to make learning Vedic Maths a structured and rewarding experience for the student.

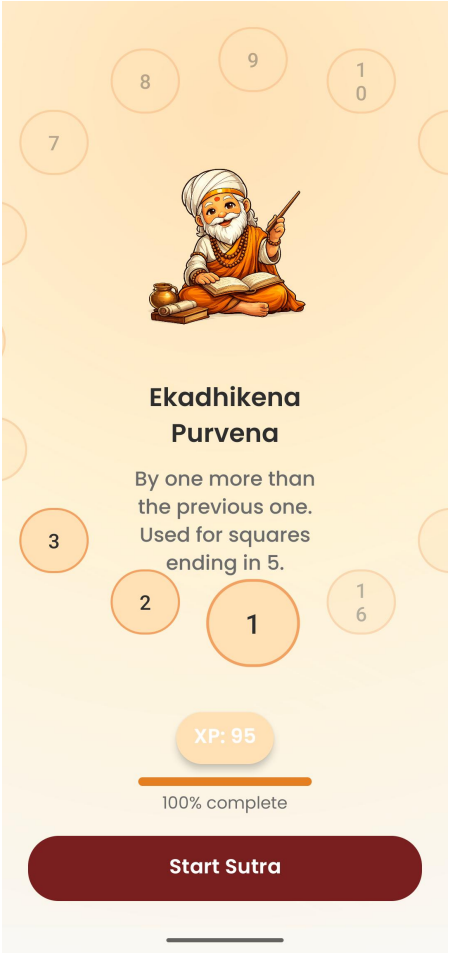


Fig. 11: Career Mode Page with Interactive 16-Sutra Dial

5.5 Challenge Mode Page

The Challenge Mode Page is the competitive and skill-based practice environment of VedaVenture, designed to strengthen arithmetic fluency, speed, and retention through interactive gameplay. While Career Mode focuses on guided conceptual learning, Challenge Mode emphasizes rapid application of previously learned Vedic Mathematics techniques in a motivating and reward-driven setting.

A major component of this mode is the **intelligent random question generation engine**. Instead of presenting a fixed question bank, the system continuously generates new arithmetic problems in real time. Each generated question is selected according to the learner's current profile, ensuring variety and preventing repetition. The generation process considers two key factors:

- **Difficulty Level:** Determines numerical complexity, number of digits, operation type, and response speed expected from the user.
- **Completed Sutras:** Questions are created only from Sutras that the learner has already unlocked and completed in Career Mode. This guarantees that users are tested only on techniques they have already studied.

For example, a beginner who has completed only the first few Sutras may receive problems involving squares ending in 5 or base-number multiplication, while advanced learners may receive mixed challenges involving division, simultaneous equations, or multi-step arithmetic shortcuts.

The mode is fully **time-based**, making speed an essential part of gameplay. A countdown timer is displayed at the top of the screen, and users must answer as many questions as possible before time expires. This timed structure improves quick decision-making, mental agility, and confidence under pressure, which is especially valuable for aptitude tests and competitive examinations.

The top section of the interface also displays important live metrics such as:

- Remaining Time
- Current Difficulty Tier
- Active Session Score
- Consecutive Correct Streak
- Earned XP

One of the most innovative features of this mode is the **joystick-based answer mechanism**. Instead of typing numerical answers manually, users interact with a custom circular controller positioned at the center of the screen. Multiple answer choices are placed around the ring, and the learner simply slides the joystick toward the correct option. This makes gameplay faster, more intuitive, and significantly more enjoyable on mobile devices.

The challenge flow is as follows:

1. A random question is generated based on unlocked Sutras and difficulty level.
2. Four answer options are displayed around the joystick dial.
3. The user slides the joystick toward the chosen answer.
4. Correct answers immediately increase score and XP.
5. Consecutive correct responses build the streak multiplier.
6. A new question appears instantly until the timer ends.

The reward system is designed to encourage consistent practice habits. Users earn **Experience Points (XP)** based on accuracy, speed of response, and streak length. Daily participation helps maintain and extend long-term streaks, reinforcing regular engagement with mathematics practice.

Incorrect answers do not terminate the session; instead, the system records performance data and continues generating new questions. This ensures a low-pressure learning environment where mistakes become opportunities for improvement rather than punishment.

As users improve over time, the adaptive engine can automatically increase challenge intensity by reducing answer time, increasing digit size, combining multiple

Sutras, or introducing more complex arithmetic patterns. Conversely, if repeated errors are detected, the system may temporarily reduce difficulty to preserve motivation and confidence.

Overall, the Challenge Mode transforms arithmetic revision into an engaging timed game where learners sharpen mental speed, gain rewards, maintain streaks, and continuously improve through unlimited randomized practice.

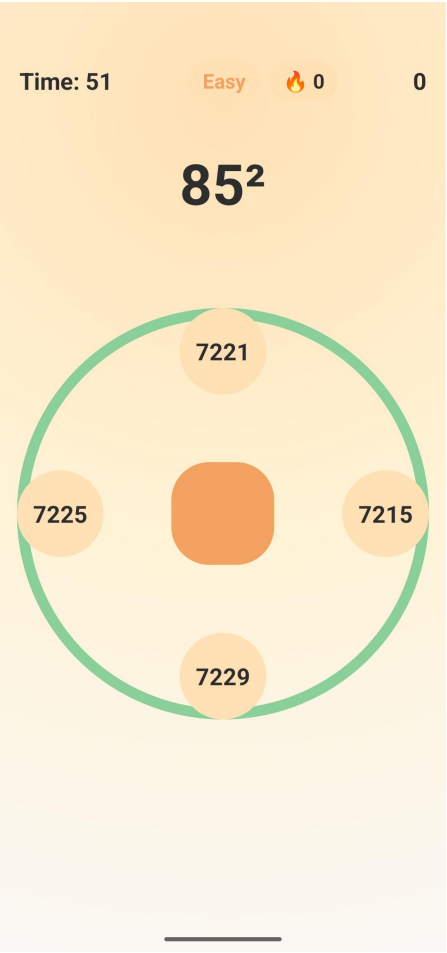


Fig. 12: Challenge Mode with Timed Randomized Questions and Joystick-Based Answer Selection

5.6 Career Mode: Interactive Sutra Learning Module

When a learner selects any unlocked Sutra from the Career Mode dial, the application opens a dedicated interactive learning module designed to teach that technique in

a highly structured, guided, and beginner-friendly manner. This module transforms traditional mathematical rules into an engaging digital lesson where users learn by understanding, observing, and practicing step by step.

At the top of the screen, the selected Sutra name is displayed prominently along with a progress bar indicating completion status of that module. This helps learners track how much of the lesson has been completed and motivates them to finish the training sequence.

Two contextual support icons are also provided:

- **Information Icon:** Displays the meaning of the Sutra, where it is applicable, and the general method used to solve problems with that technique.
- **Help / Question Icon:** Opens a worked demonstration of the current problem, showing the complete solution process before the learner attempts similar questions independently.

The core of the page follows a **guided instructional workflow**, where each Sutra is taught through progressive learning stages rather than a single explanation. This improves comprehension and reduces cognitive overload.

5.6.1 Step-by-Step Guided Learning Process

Each problem is broken into smaller logical steps and presented one at a time. Instead of asking the learner to solve the entire question immediately, the system focuses attention on a single operation at each stage.

For example, in the case of squaring numbers ending in 5 using the *Ekadhikena Purvena* Sutra:

1. Identify the digit(s) before 5.
2. Multiply that number by its successor.
3. Append 25 to the result.
4. Combine both parts to obtain the final answer.

Each step appears individually on the screen with a dedicated prompt. The learner enters the intermediate result in the answer field and presses **Check Step**.

5.6.2 Instant Feedback Mechanism

After every response, the system immediately evaluates the learner's input and provides visual feedback:

- **Correct Intermediate Step:** A positive message such as "Correct! Now for the next step." is shown, and the lesson advances.
- **Correct Final Answer:** A completion message such as "Brilliant! Final Answer Correct." is displayed.
- **Incorrect Response:** The learner is encouraged to retry, reinforcing active participation rather than passive memorization.

This continuous feedback loop helps users identify mistakes early and learn the logic behind each stage of the solution.

5.6.3 Demo Problems and Practice Transition

Before independent practice begins, the application includes guided demo problems (for example, Demo Problem 1/3) where learners are shown how to apply the Sutra in real examples. These demonstrations bridge the gap between theory and practice.

Once the learner understands the process, the module transitions to hands-on practice through the **Let's Practice** option. Practice questions then require the learner to apply the same method independently.

5.6.4 Educational Benefits

This teaching methodology offers several learning advantages:

- Breaks complex arithmetic into manageable steps.
- Encourages conceptual understanding instead of rote memorization.
- Builds learner confidence through guided success.
- Provides immediate correction and reinforcement.
- Makes ancient Vedic techniques accessible to modern mobile learners.
- Supports self-paced learning without teacher supervision.

Because all progress is stored using Cloud Firestore, completed lessons, solved exercises, and unlocked modules remain synchronized across sessions.

Overall, the Career Mode learning module serves as the instructional backbone of VedaVenture by converting each Sutra into an interactive tutor that teaches, tests, and motivates learners through a structured digital experience.

The images below show a general instructional delivery procedure for learning one Vedic Sutra (the sutras are shown on the first page of the workflow). Each further outlines how the 16 different Vedic Sutras have been covered according to similar specifications for students by using similar instructional frameworks to deliver their instruction. All 16 Vedic Sutras include a conceptual and step-by-step explanation (for the student) of the Sutra, a guided demonstration of the Sutra, practice problems, a system for providing feedback, and a way to unlock and continue to learn new concepts based on previous learning (progression). This, when all combined, provides an entirely unified and complete Vedic Mathematics curriculum on a single app.

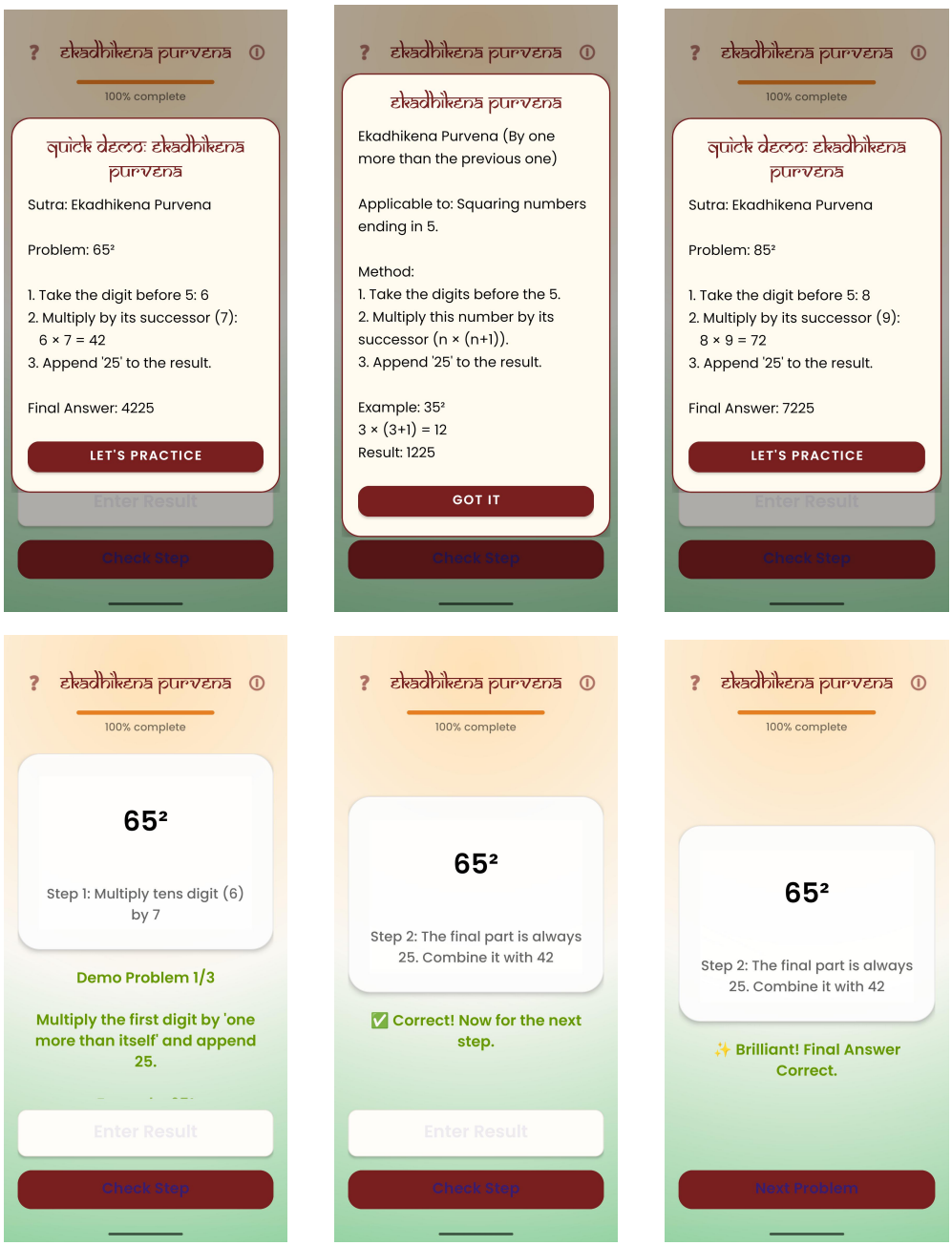


Fig. 13: Step-by-Step Guided Learning Flow in Career Mode

5.7 OCR-Based Problem Detection and Solving Module

One of the most innovative features of VedaVenture is the **OCR-Based Problem Detection and Solving Module**, which allows users to scan real-world arithmetic questions using their device camera and instantly receive the relevant Vedic Mathematics method and solution. This feature bridges the gap between classroom problems, textbook exercises, and the digital learning environment by enabling learners to apply Vedic techniques beyond pre-designed app exercises.

The module is implemented using the **Google ML Kit Text Recognition Toolkit**, which provides on-device optical character recognition capabilities for fast and reliable text extraction. By performing recognition locally on the mobile device, the system reduces latency, improves privacy, and enables smoother real-time interaction without depending entirely on cloud processing.

5.7.1 Working Procedure

The OCR workflow is divided into multiple stages:

1. **Camera Capture:** The user opens the OCR Scanner page and points the camera toward a mathematical expression printed on paper, displayed on a screen, or handwritten clearly.
2. **Region Detection:** The scanner identifies the visible problem area and highlights the detected expression using a bounding box overlay.
3. **Text Recognition:** ML Kit extracts the characters, symbols, and numbers from the image and converts them into machine-readable text.
4. **Expression Parsing:** The recognized text is analyzed to identify operators such as addition, subtraction, multiplication, powers, or division.
5. **Sutra Mapping Engine:** Based on the structure of the problem, the system predicts the most suitable Vedic Sutra for solving it.
6. **Interactive Solution Output:** The learner is asked whether they would like to solve the problem using the suggested Sutra. If accepted, the app displays the answer and explains the step-by-step method.

5.7.2 Automatic Sutra Detection Logic

The OCR module does not merely read numbers; it also performs rule-based classification to connect detected problems with learned Sutras. Examples include:

- Numbers ending in 5 with square notation (e.g., 85^2) $\rightarrow$ *Ekadhikena Purvena*
- Multiplication of general multi-digit numbers (e.g., 3×55) $\rightarrow$ *Urdhva Tiryakbhyam*
- Numbers close to powers of 10 (e.g., 98×97) $\rightarrow$ *Nikhilam*
- Division or transposition patterns $\rightarrow$ *Paravartya Yojayet*

This intelligent mapping allows learners to understand not only the answer, but also *which* Vedic technique should be applied in different scenarios.

5.7.3 Educational Benefits

The OCR module significantly extends the usefulness of the platform by allowing spontaneous practice from external sources. Instead of manually entering every question, users can simply scan and learn instantly.

Key benefits include:

- Converts textbook and worksheet questions into interactive learning opportunities.
- Helps learners identify the correct Sutra for unfamiliar problems.
- Provides immediate worked solutions and conceptual guidance.
- Encourages real-world application of Vedic Mathematics outside the app.
- Reduces typing effort and improves accessibility on mobile devices.
- Creates a smart tutoring experience using machine learning assistance.

5.7.4 Technical Advantages of ML Kit Integration

Using Google ML Kit provides several implementation advantages:

- Fast on-device inference with low latency.
- Works offline for many recognition tasks.
- Supports camera stream integration in Android applications.
- Reliable recognition of printed numeric expressions.
- Easy integration with Java/Kotlin Android projects.

Overall, the OCR Problem Detection and Solving Module transforms the smartphone camera into an intelligent mathematics assistant, enabling users to scan, understand, and solve arithmetic problems through the power of Vedic Mathematics and machine learning.

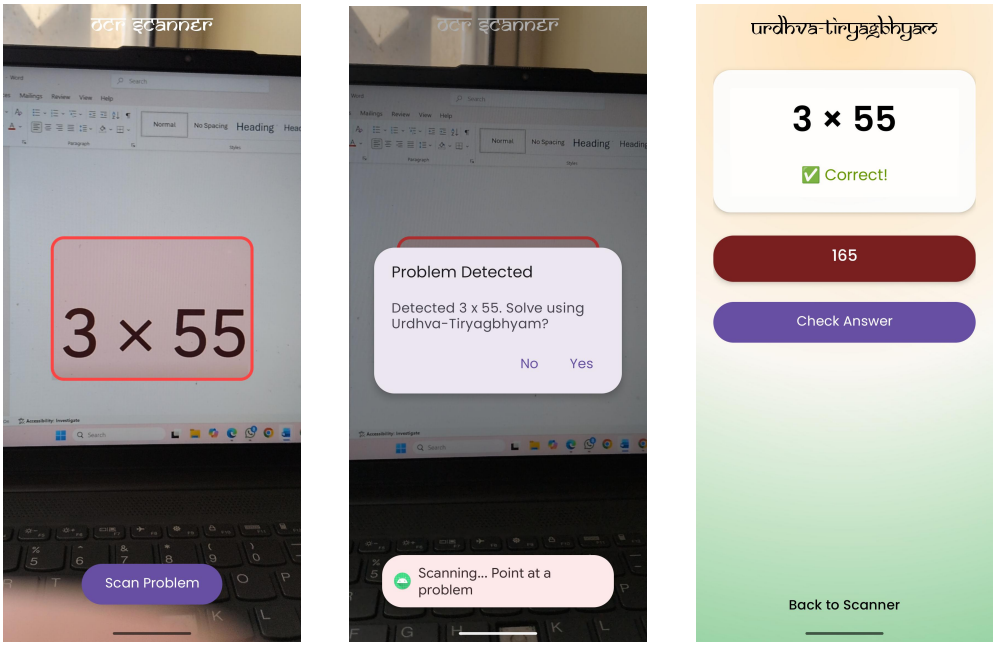


Fig. 14: OCR Scanner: Problem Detection, Recognition, and Solution Output

6 Software Requirement

6.1 Functional Requirements

- The system shall allow new users to register with a valid email and password.
- The system shall authenticate registered users and grant access to the application upon successful login.
- The system shall display a Career Mode with a list of Vedic Sutras, indicating locked and unlocked status for each.
- The system shall allow users to unlock the next Sutra only after successfully completing the assessment phase of the current Sutra.
- The system shall present step-by-step instructional content for each Sutra in Career Mode.
- The system shall present practice questions for each Sutra, with a hint system available for each question.
- The system shall provide a Challenge Mode with time-bound, dynamically generated Vedic arithmetic questions.
- The system shall implement a joystick-based directional input for answering Challenge Mode questions.
- The system shall award XP to users based on correctness, speed, and difficulty of answers.
- The system shall maintain and display a daily streak counter, resetting after 24 hours of inactivity.

- The system shall record and display the user's best streak.
- The system shall adapt question difficulty in Challenge Mode based on user accuracy and response time.
- The system shall provide an OCR scanner that captures an arithmetic question from an image and returns the relevant Vedic Sutra and step-by-step solution.
- The system shall persist all user data, including XP, streaks, Sutra progress, and difficulty level, to Firebase Firestore.
- The system shall synchronize user data in real time across devices.

6.2 Non-Functional Requirements

- **Performance:** The application shall load the Dashboard within 2 seconds on a standard 4G connection.
- **Reliability:** The Firebase backend shall maintain 99.9% uptime availability.
- **Security:** All user credentials shall be managed through Firebase Authentication with encrypted storage.
- **Usability:** The user interface shall conform to Google Material Design guidelines and support screen sizes from 5 to 7 inches.
- **Scalability:** The Firebase Firestore database shall support up to 100,000 concurrent users without degradation of performance.
- **Maintainability:** The application codebase shall be modular with clearly separated layers for presentation, logic, and data access.
- **Accessibility:** Text sizes shall meet WCAG minimum contrast and size requirements for mobile displays.

7 Conclusion

The Veda Venture application successfully addresses the lack of motivation and engagement in traditional arithmetic learning. By delivering Vedic Mathematics through a gamified, adaptive mobile platform, the app improves both user motivation and computational proficiency. The integration of structured Career learning, adaptive Challenge testing, and real-time OCR problem-solving creates a comprehensive ecosystem for mathematical training. Future iterations of the application could focus on expanding the OCR capabilities to handle complex algebraic equations, incorporating advanced data analytics to track user pain points, and introducing a multiplayer leaderboard system to further leverage the gamification elements.

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